PHYS 357: Honours Quantum Physics 1

W.R.YCH.

August 31^{st}, 2026

Abstract

NOT FINISHED

Contents

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 The classical picture

2.1.1 Angular momentum

Definition 1 (Orbital angular momentum)**.** *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \tag{1}$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x-y plane, only the z component survives. On a circular path of radius r with speed v,

$$L_z = mvr.$$

Definition 2 (Angular momentum of a rigid body)**.** *For a rigid body*[1] *rotating about a fixed axis this is*

$$L = I_0\omega, \tag{2}$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 1. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

2.1.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment)**.** *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

[1] A rigid body is one that does not deform under the action of external forces.

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c}[2] = \frac{1}{c}\left(\frac{qv}{2\pi r}\right)\left(\pi r^2\right) = \frac{q}{2mc}\,mvr = \frac{q}{2mc}\,L_z, \tag{3}$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

2.1.3 Gyromagnetic ratio

Definition 4 (Gyromagnetic ratio)**.** *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \tag{4}$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \tag{5}$$

Remark 2. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

2.2 Spin

Definition 5 (Spin)**.** *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

2.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \qquad \gamma = \frac{qg}{2mc}, \tag{6}$$

where g is the g-factor of the particle.

Definition 6 (g-factor)**.** *The g-factor of a particle[3] is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

[2] Here c is in Gaussian units.

[3] Its value must be measured or derived from relativistic quantum theory.

2.2.2 Magnetons

Definition 7 (Planck constant)**.** *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s.} \tag{7}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 8 (Reduced Planck constant)**.**

$$\hbar = \frac{h}{2\pi}. \tag{8}$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{kg \cdot m^2}{s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 9 (Bohr magneton)**.** *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \tag{9}$$

Definition 10 (Nuclear magneton)**.** *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \tag{10}$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 11 (Nucleons)**.** *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \, \mu_B \, \frac{\vec{S}}{\hbar}, \tag{11}$$

and similarly for the proton and neutron with μ_N in place of μ_B.

2.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g-factor	Gyromagnetic ratio γ
Electron	$-\lvert e\rvert$	m_e	$g_e \approx 2$	$g_e\mu_B/\hbar$
Proton	$+\lvert e\rvert$	m_p	$g_p \approx 5.59$	$g_p\mu_N/\hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n\mu_N/\hbar$

Table 1: Charges, masses, g-factors and gyromagnetic ratios.

Remark 3. *The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 4. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

2.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach apparatus)**.** *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z. The gradient deflects each atom by an amount proportional to μ_z, and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z.*

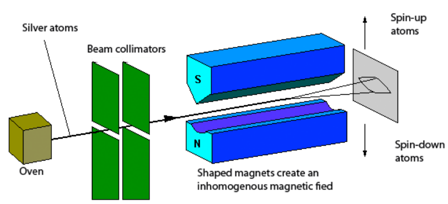

Figure 1: Illustration of the Stern-Gerlach apparatus.

2.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \tag{12}$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U, \qquad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}. \tag{13}$$

Definition 13 (Deflection)**.** *The transverse displacement of an atom on the screen, produced by the force F_z acting over the length of the magnet. It is proportional to μ_z, so the pattern on the screen is a record of the moments in the beam.*

Remark 5. *A uniform field gives no force, only a torque. The apparatus needs a field* gradient *to deflect anything.*

2.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e|e|}{2m_e c} S_z, \tag{14}$$

and therefore

$$F_z = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z, \tag{15}$$

where the last equality assumes $\partial B_z/\partial z < 0$.

Result 1.

$$F_z \propto S_z. \tag{16}$$

The deflection of an atom measures the z component of its spin directly.

2.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \qquad S_z = |\vec{S}| \cos\theta. \tag{17}$$

Classically $\theta \in [0, \pi]$, so $\cos\theta \in [-1, 1]$ and S_z varies continuously over $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, the beam should smear into a continuous band. It does not.

2.4 Quantization of S_z

Result 2 (Quantization of spin)**.** *The electron spin angular momentum is quantized:*

$$S_z = \pm\frac{\hbar}{2}. \tag{18}$$

The beam splits into two spots.

2.4.1 Ket notation

New notation, a "ket":

$$|{\uparrow}\rangle \text{ for } S_z = +\frac{\hbar}{2}, \qquad |{\downarrow}\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The spin of an electron is a two-level system, so it can be represented as a two-dimensional vector space. The spin state of an electron can be represented as a linear combination of the basis states $|{\uparrow}\rangle$ and $|{\downarrow}\rangle$:

A Appendix

B Useful Links

Stern-Gerlach experiment: https://www.youtube.com/shorts/QG7kJXxN1Kk